

Beyond Aggregate Reliability in ADMET Prediction: Protecting Weakest-Class Coverage under Molecular Distribution Shift with a Transport-Audited Multi-View Envelope

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Short title: Auditable ADMET Reliability under Molecular Shift

Abstract

Aggregate accuracy and marginal coverage can conceal outcome-class failure and uninformative uncertainty under molecular distribution shift. We conducted a two-stage, representation-controlled study. First, frozen ECFP predictors for BBBP, ClinTox, ESOL, and Lipophilicity were audited under repeated random, label-blind scaffold, and similarity-cluster splits using point-performance, calibration, applicability-domain, conformal, and composition-aware selective diagnostics. Random splitting was generally optimistic, while chemical similarity ranked risk inconsistently across endpoints. At nominal 90% coverage, ClinTox marginal and shift-weighted sets achieved overall coverage near 0.90 but only 0.068–0.158 positive-class coverage. Mondrian calibration raised positive coverage to 0.926–0.940, but returned two-label sets for about 71% of compounds. These failures motivated TAME, a transport-audited multi-view envelope evaluated under the separately frozen RACER-C4 protocol. Two label-free weighting views can expand an ordinary Mondrian set only after prespecified transport checks and consensus; failed audits trigger exact fallback. Baseline containment structurally prevents lower same-row coverage and newly created confident singletons, so the nontrivial test was whether a sealed external cohort showed measurable

weakest-class gain at an acceptable deferral cost. In six prespecified EPA endpoints across five fresh seeds, TAME increased mean minimum-class coverage by 1.3649 percentage points (hierarchical-bootstrap 95% interval, 0.5827–2.0051) while Macro correct-singleton yield decreased 1.6067 points, inside a prespecified 5-point efficiency guardrail. Both approved views passed all 60 endpoint–seed audits; the diagnostic ECFP view failed all 60. TAME produced no empty sets or new singletons. This locked Tox21 evidence supports auditable uncertainty expansion, not arbitrary-shift coverage or universal superiority.

Keywords: ADMET prediction; conformal prediction; applicability domain; covariate shift; Tox21; transport audit

1 Introduction

In silico prediction of absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties is now embedded in compound prioritization, where large public task collections and accessible prediction platforms make it possible to screen many endpoints before experimental follow-up^{1–6}. Message-passing networks, geometry-aware models, chemical-language representations, and quantum-informed pretraining have extended the predictive frontier^{7–10}. Yet a better point predictor does not answer a different and operationally important question: when should an individual molecular prediction be treated as informative, uncertain, or in need of deferral?

Aggregate benchmark scores provide an incomplete answer because molecular observations are not independent points in an unstructured feature space. Random splitting can place close analogues on both sides of an evaluation boundary, whereas scaffold and similarity-cluster designs probe different forms of chemical extrapolation. Both predictive performance and uncertainty-method rankings can therefore change with the endpoint, representation, and split mechanism^{11–16}. Repeated, paired evaluation is also necessary: model families are not independent statistical replicates, and a cross-model average cannot replace uncertainty across matched data partitions¹⁷.

Reliability itself is multidimensional. Probability calibration asks whether reported prob-

abilities agree with empirical frequencies¹⁸. Applicability-domain analysis asks whether a query resembles the chemical space represented in training, but molecular novelty, model uncertainty, and realized error are not interchangeable^{14;19}. Selective prediction asks whether rejecting uncertain cases lowers risk, yet in an imbalanced endpoint the same curve can improve because the rejection rule removes one class faster than another. Modern uncertainty reviews consequently distinguish uncertainty sources and require the quality of an uncertainty score to be evaluated rather than inferred from predictive accuracy^{20;21}. Molecular applications combining uncertainty with attribution or reaction-property prediction further show that useful uncertainty can be local and task specific^{22–24}.

Conformal prediction supplies prediction sets or intervals with finite-sample marginal coverage under exchangeability and has been adapted to toxicity, ADMET, graph models, and chemical covariate shift^{25–27}. Marginal validity, however, does not imply class-conditional or subgroup validity. In a rare-event classifier, an overall coverage value close to its nominal target can coexist with near-total failure for the minority class^{28–31}. Conversely, a class-conditional procedure can raise coverage by returning both labels for most compounds. Coverage and informativeness must therefore be reported together³².

Molecular distribution shift makes this balance more difficult. Weighted conformal prediction has a clear justification when a target-to-source likelihood ratio is known³³; in practice, molecular ratios are estimated from finite unlabeled samples in a chosen representation. An estimated ratio can collapse effective sample size, saturate clipping bounds, or reduce one domain discrepancy while worsening another. Adaptive conformal methods address other nonexchangeable settings^{34;35}, but algorithmic complexity alone does not establish a useful coverage–efficiency trade-off. What is missing from many empirical workflows is an explicit certificate for when a transport correction is permitted to affect the final set and an exact fallback for when that certificate fails.

Recent conformal methods pursue related but distinct objectives. KMM-CP uses kernel mean matching and support-overlap selection for covariate-shift correction, including molecular benchmarks, while multi-distribution robust conformal prediction assumes labeled data from

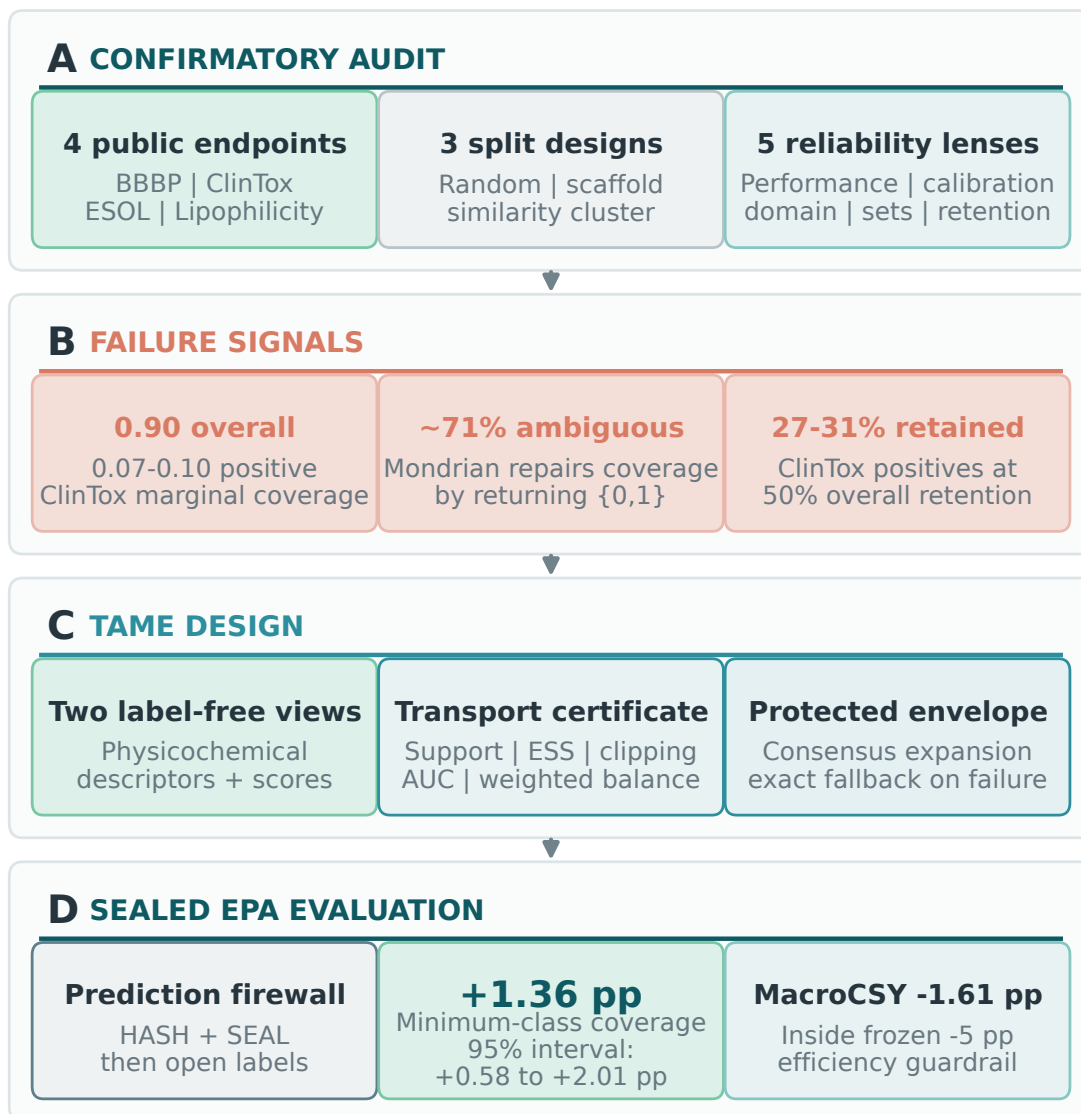
multiple source distributions and targets uniform validity across those sources or their mixtures^{36;37}. Singleton-optimized conformal prediction redesigns the score to increase singleton frequency, whereas batch conformal prediction constructs joint label-vector sets for multiple test examples^{38;39}. TAME addresses neither multi-source robustness nor joint batch labels: it produces per-compound binary sets for one unlabeled target cohort, admits transport corrections only after view-specific audits, and expands a protected baseline rather than optimizing it away. These differences make the fallback and the measured coverage–deferral trade-off central to its evaluation.

Broad contemporary ADMET benchmarks answer a complementary question by comparing foundation models, graph networks, AutoML systems, and classical predictors under data scarcity, structural out-of-distribution evaluation, class imbalance, activity cliffs, and beyond-Rule-of-5 space⁴⁰. Our focus is not another architecture leaderboard. We hold representation and predictor families fixed in the diagnostic stage, then ask whether commonly reported reliability lenses agree and whether their observed failures can be converted into design constraints for an auditable uncertainty layer.

The study has two prospectively separated stages (Figure 1). Stage I is a frozen four-endpoint audit of BBBP and ClinTox classification and ESOL and Lipophilicity regression under random, label-blind scaffold, and label-blind similarity-cluster splits. It tests four questions: how molecular distribution shift changes point performance and calibration; when chemical similarity ranks error or miscoverage; whether aggregate coverage conceals class failure; and whether conformal or selective remedies remain informative. Stage II translates the resulting constraints into the transport-audited multi-view envelope (TAME) and evaluates this method under the separately frozen RACER-C4 protocol. TAME uses an ordinary class-conditional Mondrian set as a protected baseline, two label-free transport views that must pass prespecified audits, and an all-view consensus that may add a protected label but may never remove a baseline label or create a new confident singleton.

Architecture development was restricted to the public Tox21 leaderboard cohort⁴¹. The method, primary endpoints, five fresh seeds, promotion criteria, a -5 percentage-point point-

estimate guardrail for Macro correct-singleton yield (MacroCSY), and the negative-result policy were frozen before the independent EPA labels were opened. Final predictions and transport audits were written and hashed before the label file was acquired and parsed. This ordering separates a useful empirical result from a stronger claim that the design cannot support: TAME is evaluated as an audited, failure-aware layer in a locked Tox21 setting, not as an exact conditional-coverage method or a universal solution to arbitrary molecular shift.



60/60 final endpoint-seed cells completed under the prediction-to-label firewall

Figure 1: **Evidence chain from reliability diagnosis to independent validation.** (A) A frozen four-endpoint audit compared point performance, calibration, conformal validity, chemical-domain diagnostics, and selective retention under three label-blind split designs. (B) The audit exposed failure signals that aggregate summaries concealed. (C) These signals became explicit TAME design constraints: two audited label-free transport views, a baseline-containing consensus, and exact fallback. (D) Under the RACER-C4 evaluation protocol, final EPA predictions were hashed and sealed before labels were opened; the primary result was a gain in minimum-class coverage with the MacroCSY point estimate inside the frozen efficiency guardrail.

2 Methods

2.1 Stage I: confirmatory audit and evidence boundary

The diagnostic stage was organized as a frozen confirmatory benchmark rather than a post hoc search for favorable endpoint–method combinations. Development splits and seed 99 were used only for implementation checks and exploratory debugging. Scientific conclusions were restricted to previously unseen confirmatory splits: seeds 101–110 for random and scaffold designs and seeds 101–105 for similarity-cluster designs. Model families, imbalance regimes, conformal methods, applicability-domain definitions, selective-prediction metrics, and stopping rules were fixed before confirmatory outputs were inspected. No endpoint, model, seed, threshold, or unfavorable result was removed after inspection. This separation follows recommendations that molecular machine-learning comparisons distinguish practical effects, paired evaluation units, and repeated-split uncertainty rather than treating model architectures as independent replicates¹⁷.

2.2 Endpoint acquisition and structure standardization

Four public MoleculeNet/DeepChem endpoints were selected to provide two complementary classification and regression stress-test pairs⁴². BBBP used the binary `p_np` endpoint, ClinTox used `CT_TOX`, ESOL used measured log aqueous solubility in mol L^{-1} , and Lipophilicity used the experimental `exp` target. The panel spans different sample sizes, class prevalences, and continuous-target distributions, but was not intended to represent the full diversity of ADMET endpoints. BBBP is historically associated with in silico blood–brain barrier penetration modeling⁴³, whereas ESOL originates from the Delaney aqueous-solubility collection⁴⁴.

Raw files were downloaded from the public DeepChem dataset distribution and hashed with SHA-256. Rows with missing structures or targets were removed, and SMILES strings were canonicalized with RDKit. Molecules that could not be parsed after canonicalization were discarded. Duplicate canonical structures were resolved before featurization. For classification, the majority label among duplicate records was retained; an exact tie between conflicting labels was treated as ambiguous and removed. For regression, duplicate target values were

aggregated by their median. Table 1 reports the resulting core datasets. The ClinTox positive prevalence of 6.45% was retained rather than artificially balanced outside the training-only resampling regime.

Table 1: **Core endpoint composition after structure cleaning and duplicate resolution.** Classification prevalence is the fraction labeled positive. Regression ranges are reported in the native target scale distributed with MoleculeNet/DeepChem.

Endpoint	Task	Raw n	Clean n	Removed	Conflicts	Prevalence or target range
BBBP	Classification	2050	1965	85	10	0.7634 positive
ClinTox	Classification	1484	1442	42	19	0.0645 positive
ESOL	Regression	1128	1117	11	6	-11.60 to 1.58
Lipophilicity	Regression	4200	4200	0	0	-1.50 to 4.50

2.3 Molecular representation

Canonical structures were encoded as 2048-bit binary Morgan fingerprints with radius 2, corresponding to ECFP4⁴⁵. The same frozen fingerprint matrix was used for model fitting, similarity-based splitting, density-ratio estimation, and applicability-domain analysis. This representation-controlled design was chosen to isolate reliability diagnostics from changes in learned representation and was not intended as an architecture leaderboard. Modern graph, geometry-aware, and foundation-model encoders may change predictive performance and uncertainty behavior, but varying the representation within this study would confound whether a difference arose from the diagnostic or from the molecular feature space itself^{7;8;10;40}.

2.4 Random, scaffold, and similarity-cluster splits

Every confirmatory split assigned approximately 60% of molecules to training, 20% to calibration, and 20% to testing. Models and training-only resampling used the training role. Calibration, conformal thresholds, density-ratio estimation, and learned error-scale transformations used only training and/or calibration information as specified below. Test labels were never used for model fitting, split construction, threshold selection, or method selection.

Random classification splits were generated in two stages with stratification whenever both classes had sufficient support; regression random splits were unstratified. The same seed

controlled the train-versus-temporary split and the equal calibration-versus-test partition of the temporary set.

For scaffold evaluation, Bemis–Murcko scaffold groups already stored in the processed split files were assigned intact and without target information. Scaffold groups were first shuffled by the prespecified seed, stably ordered by decreasing group size, and allocated to the role with the smallest projected fill ratio relative to its target size. Seed-derived tie ordering was used only when projected ratios were equal. Structural validation required every molecule to receive one role and every scaffold to occur in exactly one role.

Similarity-cluster evaluation used an order-randomized leader algorithm on binary ECFP4 fingerprints. This was not exact Butina clustering. For each seed, molecules were visited in a seeded random order; a molecule joined the most similar existing leader when its Tanimoto similarity was at least 0.60 and otherwise initiated a new cluster. Clusters were then assigned intact to training, calibration, or testing using group size only and the same projected-fill principle. The clustering step was prespecified for endpoints with at most 10,000 molecules; all four core endpoints satisfied this cap. Validation required zero cluster crossover between roles.

2.5 Frozen predictive models and imbalance regimes

Four model families were evaluated for each endpoint. Classification used logistic regression, random forest, XGBoost, and a multilayer perceptron on ECFP4. Regression used ridge regression and the corresponding random-forest, XGBoost, and ECFP multilayer-perceptron regressors.

The linear classification pipeline used sparse-compatible standardization and logistic regression with the liblinear solver, 2000 maximum iterations, no class weighting, and the split seed as the random state. Ridge regression used sparse-compatible standardization and ℓ_2 penalty $\alpha = 1.0$. Random forests used 500 trees and the split seed. XGBoost used 300 estimators, maximum depth 4, learning rate 0.05, subsampling 0.90, column subsampling 0.90, histogram tree construction, and the split seed. The multilayer perceptron used sparse-compatible standardization, hidden layers of 256 and 128 units, ReLU activation, ℓ_2 coefficient

10^{-4} , a maximum of 300 iterations, early stopping, and the split seed.

Classification was evaluated under two separate regimes. The unweighted regime trained every model on the original training distribution. In the balanced-oversample regime, the minority class was sampled with replacement within the training role until it matched the majority count; the same seeded resampled training matrix was supplied to all four model families. Calibration and test roles were never resampled. Regression required no imbalance regime. Architecture comparisons never mixed weighted and unweighted training as though they were a single model ranking.

2.6 Predictive performance and probability calibration

Classification performance was summarized by ROC-AUC, PR-AUC, and balanced accuracy; regression performance used RMSE, MAE, and R^2 . For binary probability evaluation, predictions were clipped to $[10^{-7}, 1 - 10^{-7}]$ before the Brier score and negative log-likelihood were calculated. Calibration was examined in two distinct ten-bin views¹⁸: positive-probability calibration compared mean $P(Y = 1)$ with observed positive frequency, whereas confidence calibration compared maximum class probability with empirical correctness. Expected calibration error was the bin-frequency-weighted absolute gap, and maximum calibration error was the largest occupied-bin gap. These two views were retained separately because calibration of the positive probability is not equivalent to calibration of classification confidence.

2.7 Marginal and Mondrian conformal prediction

All headline conformal analyses used $\alpha = 0.10$. For binary classification, the least-ambiguous-classification nonconformity score was

$$s_i = 1 - \hat{p}_{y_i}(x_i), \tag{1}$$

where $\hat{p}_{y_i}(x_i)$ is the probability assigned to the observed class. Marginal split conformal prediction used one calibration distribution for both labels. Its finite-sample order statistic

was indexed by

$$k = \min\{n_{\text{cal}}, \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil\}. \quad (2)$$

A candidate label was included when its test nonconformity score did not exceed the corresponding threshold. Mondrian prediction used separate calibration-score distributions and thresholds for class 0 and class 1, providing class-conditional rather than only marginal calibration motivation^{29;31}. We reported overall and class-specific coverage, the absolute class-coverage gap, prediction-set size, empty-set rate, singleton rate, and ambiguous two-label rate. These efficiency measures were considered jointly with validity because class-conditional coverage can be repaired by returning weakly informative sets^{28;32}.

For regression, standard split conformal intervals used absolute calibration residuals $|y_i - \hat{y}_i|$ and the same finite-sample quantile rule. Test intervals were symmetric around the point prediction.

2.8 Empirical covariate-shift-weighted conformal baseline

The shift-aware baseline followed the weighted-conformal principle that target-domain coverage can be recovered under covariate shift when the relevant likelihood ratio is known or estimated sufficiently accurately³³. In each run, calibration ECFP4 vectors were assigned domain label 0 and the unlabeled test ECFP4 vectors domain label 1. Density ratios were estimated with five-fold cross-fitted ℓ_2 logistic regression using $C = 1.0$, liblinear optimization, and no class weighting. Domain-classifier odds were multiplied by $n_{\text{cal}}/n_{\text{test}}$ and clipped to $[0.05, 20]$. Weighted conformal quantiles included the test-point mass at infinity. We recorded domain-classifier AUC, effective sample size of the calibration weights, infinite-threshold frequency, empirical coverage, and efficiency.

Because the ratios were estimated from finite molecular samples rather than known analytically, these results are described as empirical shift-aware diagnostics, not as exact guarantees under arbitrary molecular distribution shift. CoDrug provides a recent molecular example of density-aware conformal correction under covariate shift, but uses a different density-estimation strategy²⁷.

2.9 Adaptive normalized regression intervals

The locally adaptive regression baseline divided the calibration role deterministically into equal scale-fitting and conformal-quantile halves using a random generator derived from the split seed. A random-forest scale model with 300 trees, `min_samples_leaf=20`, and `max_features=sqrt` was fitted on the first half to predict $\log(1 + |y - \hat{y}|)$. The second half supplied normalized residual scores. The predicted scale was floored at

$$\max\{0.1 \text{ median}(|r|_{\text{scale}}), 0.001\}, \quad (3)$$

and the test half-width was the normalized conformal quantile multiplied by the predicted test scale. Evaluation included coverage, absolute coverage gap, interval width, width coefficient of variation, infinite-interval rate, and Pearson and Spearman association between width and realized absolute error.

2.10 Applicability-domain diagnostics

For each test molecule, the primary chemical-domain score was its maximum Tanimoto similarity to the training fingerprints. Mean top-five similarity and unseen-scaffold status were also generated in the pipeline, while the main manuscript emphasizes maximum similarity because it was used consistently in continuous, threshold, and selective analyses. The complementary out-of-domain score was $1 - \text{maximum similarity}$. We calculated Spearman association between similarity and predictive risk and between similarity and marginal-conformal miscoverage. Out-of-domain AUC measured whether the complementary score discriminated erroneous or miscovered observations.

Prespecified descriptive bins were low domain (< 0.40), medium domain ($0.40 - < 0.70$), and high domain (≥ 0.70). Because any fixed structural-similarity threshold is representation and endpoint dependent¹⁹, primary inference used continuous association. Sensitivity analyses swept thresholds from 0.30 to 0.80 and also examined similarity quartiles. Sparse or single-class groups were retained with warnings rather than silently excluded.

2.11 Selective prediction and retained-class diagnostics

Classification samples were ranked separately by 1–maximum class probability and by 1–maximum Tanimoto similarity; regression used the latter. Retained coverage ranged from 0.10 to 1.00 in increments of 0.05. At every retained fraction, the selected subset was compared with 500 deterministic repeated random-rejection samples of the same size. Classification outputs included ordinary error, balanced error when both classes remained, false-negative and false-positive rates, positive and negative retention, and the shift in retained class balance. Regression outputs included RMSE and MAE.

The area metric was explicitly reported as partial AURC because integration covered retained fractions 0.10–1.00 rather than the full mathematical domain. Selective and random balanced-error curves were integrated only over matched coverage points for which both quantities were finite. This matched-support rule prevents apparent improvements caused solely by dropping different undefined low-coverage points.

2.12 Aggregation, paired comparisons, and uncertainty

Metrics were first summarized within a fixed endpoint, split type, seed, model, and imbalance regime. Model-specific means, standard deviations, and two-sided 95% Student-*t* intervals were then calculated across confirmatory seeds. Method contrasts were paired within the same endpoint, split, seed, model, and regime. Cross-model headline averages were used only as compact descriptive summaries; model families were not treated as independent inferential replicates. For each paired effect, we additionally counted how many frozen model or model–regime combinations had a model-specific confidence interval entirely above or below zero. No multiplicity-adjusted null-hypothesis testing across model families was used, and null or adverse effects were retained.

2.13 Stage II: Tox21 cohorts and prospective roles

The design stage used the official Tox21 Challenge cohorts, which comprise nuclear-receptor and stress-response assays measured on environmental chemicals and drugs⁴¹. The Tox21

10K training collection supplied open labels for model fitting, routing, and internal conformal calibration. A public leaderboard batch of 296 structures supplied architecture-development labels. A separate official file of 647 final structures was the unlabeled target for one independent evaluation. All URLs, archive members, SHA-256 hashes, endpoint order, preprocessing rules, seeds, audit thresholds, promotion gates, and failure actions were recorded in an executable lock before final-label access.

Six primary endpoints—NR-AhR, NR-ER, SR-ARE, SR-ATAD5, SR-MMP, and SR-p53—were selected by the frozen rule of at least 20 positive labels in the public development batch, not by TAME performance. The six remaining official assays were retained as secondary small-sample stress tests and were never used to enlarge the primary denominator. Development used seeds 101–105; independent evaluation used fresh seeds 211–215. Standardized structures overlapping the 10K training cohort were excluded from external domain fitting and endpoint evaluation. An unparseable external structure retained its public identity, received the non-actionable full set $\{0, 1\}$, and was excluded with an explicit reason code.

2.14 Frozen base predictor and ordinary Mondrian set

For each endpoint and seed, the Tox21 training cohort was partitioned by a stratified, seed-fixed 60/20/20 allocation into fit, router, and internal conformal roles. Four ECFP4 classifiers were fitted on the fit role: logistic regression, random forest, extra trees, and Bernoulli naive Bayes. The tree ensembles used 120 estimators and square-root feature sampling; the two linear-probabilistic models and all remaining hyperparameters were fixed in the executable lock. An ℓ_2 logistic router was trained on the four component logits in the router role.

For routed probability $p(x)$, the candidate-label nonconformity scores were $s_0(x) = p(x)$ and $s_1(x) = 1 - p(x)$. Ordinary class-conditional Mondrian thresholds at $\alpha_0 = \alpha_1 = 0.10$ formed the protected baseline $S_M(x)$. Within class y , the finite-sample rank was

$$k_y = \lceil (n_y + 1)(1 - \alpha_y) \rceil, \quad (4)$$

with an unavailable rank mapped to positive infinity. This base layer was fixed before either

external cohort was evaluated.

2.15 Label-free transport views and audit certificate

TAME estimated source-to-target density ratios from two complementary views that used external structures and model outputs but no external labels. The physicochemical view contained molecular weight, logP, topological polar surface area, hydrogen-bond donor and acceptor counts, rotatable-bond count, ring count, fraction sp^3 , and heavy-atom count. The score view contained the four component logits and the routed logit. A 2048-bit ECFP domain classifier was retained as a mandatory diagnostic comparator but was not eligible to enter the envelope after failing its development certificate.

Each view used five-fold cross-fitted ℓ_2 logistic domain classification for the domain-AUC audit and the same regularized estimator refitted on all unlabeled source and target covariates for the working density ratio. Posterior odds were prior-corrected by $n_{\text{source}}/n_{\text{target}}$ and clipped to $[0.05, 20]$. The weighted source scores and query mass at positive infinity then defined candidate-specific weighted Mondrian thresholds^{27;33}. Unlike KMM-CP, which estimates covariate-shift weights through constrained kernel mean matching and adds a support-overlap selection step³⁶, TAME retained the frozen logistic ratio estimator and used the complete certificate below as an activation gate. KMM-CP was not part of the locked comparator suite; adding a new final-label comparison after unsealing would violate the evidence boundary, so a head-to-head test requires a new prospectively locked study.

A view was active only if all six prespecified checks passed: (i) at least 100 source and 100 target structures; (ii) at least 20 internal conformal examples in each true class; (iii) total and per-class effective sample size at least 25% of the corresponding raw count; (iv) no more than 30% of weights at either clipping boundary; (v) cross-fitted domain AUC no greater than 0.99; and (vi) the weighted standardized mean gap no more than 1.05 times the unweighted gap. These diagnostics are empirical transport certificates. They do not prove that estimated ratios are correct or restore exact coverage under arbitrary molecular distribution shift.

2.16 TAME envelope, fallback, and structural invariants

Public development showed a mean coverage deficit for label 0 greater than two percentage points and no mean deficit for label 1. The protected label was therefore frozen as label 0 before the final EPA evaluation. This label choice used public development outcomes; only the application of the two transport views to the final target was label free.

Let $\bar{S}_M(x) = S_M(x)$ when the baseline is nonempty and $\bar{S}_M(x) = \{0, 1\}$ otherwise. Let $S_{\text{phys}}(x)$ and $S_{\text{score}}(x)$ be the two weighted Mondrian sets. If both views pass their complete audits,

$$S_{\text{TAME}}(x) = \bar{S}_M(x) \cup \{0 : 0 \in S_{\text{phys}}(x) \cap S_{\text{score}}(x)\}. \quad (5)$$

If either approved view fails, every transport augmentation is disabled and $S_{\text{TAME}}(x) = \bar{S}_M(x)$. Thus a failed audit is observable and produces an exact ordinary-baseline fallback rather than a silently relaxed certificate.

Equation 5 yields three target-label-independent invariants: the TAME set contains the ordinary Mondrian set, TAME emits no empty set, and every TAME singleton was already a baseline singleton. Consequently, empirical coverage for either class cannot decrease on the same rows, and wrong-singleton exposure cannot increase. TAME can convert a baseline singleton into an explicit two-label defer, but cannot create a new confident singleton. Correct-singleton yield can therefore decrease but cannot increase; this motivated a prespecified efficiency guardrail rather than an efficiency-superiority claim.

2.17 Development gate and prediction-to-label firewall

The public development denominator was fixed at six primary endpoints by five seeds (30 endpoint–seed cells). Promotion required all cells to complete, zero baseline-inclusion violations, zero TAME empty sets, zero newly created singletons, nonnegative mean coverage change for both classes, nonpositive wrong-singleton-exposure change for both classes, mean MacroCSY change of at least -0.05 , and both approved views active in at least 80% of cells. Failure of any check would stop execution before final-label download or parsing, and the failed gate would remain recorded.

After the gate passed, fresh-seed models generated every comparator and TAME prediction from the 647 final structures. The sealed prediction file, final transport-audit table, and executable lock were hashed into a promotion record with `final_labels_opened=false`. Only then could the locked final-label bytes be acquired, their hash verified, and labels identity-joined for a single evaluation. The parser had no callable path around the promotion record, and contract tests exercised missing or invalid records. No endpoint, seed, transport view, threshold, protected label, or margin could be changed after final-label inspection.

2.18 Primary estimand, efficiency metric, and inference

For class y , coverage was the fraction of true- y compounds whose set contained y . The primary endpoint-level quantity was the smaller of the two class coverages. Correct-singleton yield for class y was the fraction of true- y compounds receiving the singleton $\{y\}$; MacroCSY was the unweighted mean of the two class-specific yields. Wrong-singleton exposure was analogously the fraction receiving the singleton containing the incorrect label. Ambiguous and empty rates completed the descriptive set metrics.

The primary estimand was the endpoint–seed-equal mean change in minimum-class coverage for TAME versus the ordinary routed Mondrian baseline across the six primary endpoints and five independent seeds. Its 95% interval used a frozen hierarchical bootstrap: endpoints were resampled first; within every drawn endpoint, labeled final compounds were then resampled separately within true class, preserving class counts; all five seeds were retained. The procedure used 2000 draws and seed 44021. Secondary estimands included MacroCSY change, class-specific coverage, wrong-singleton exposure, ambiguity, and empty-set rate. Interpretation was hierarchical: a positive primary point estimate had to be accompanied by a MacroCSY point estimate no worse than the frozen -5 -point guardrail. On the class-balanced MacroCSY scale, this threshold permits at most five fewer correct singleton outputs per 100 cases. It was fixed after public development as an architecture-selection tolerance and before final-label access; it was not calibrated to clinical or regulatory utility. The executable lock names the field a non-inferiority margin, but no confidence-interval-based non-inferiority test for MacroCSY was

prespecified. We therefore treat it as a point-estimate efficiency guardrail and make no formal non-inferiority claim. No multiplicity-adjusted secondary superiority claim was authorized automatically.

A post-evaluation implementation audit found that the first bootstrap report traversed unique resampled endpoints through an unordered Python set. This affected only how a fixed random stream was assigned within the percentile calculation. A hash-verified inference-only repair preserved first-occurrence endpoint order and recomputed the interval without refitting a model, regenerating a prediction, changing a label, or altering either point estimate. The repaired deterministic interval is the publication-facing value reported here; the original sealed report remains versioned in the audit chain.

3 Results

Unless stated otherwise, Stage I values are descriptive means across frozen model or model–regime combinations; model families were not treated as independent inferential replicates. Audit contrasts were paired on the same endpoint, split, seed, model, and regime, and seed 99 was excluded from scientific conclusions. Stage II development and final results are reported separately so that architecture-selection evidence is not confused with independent validation.

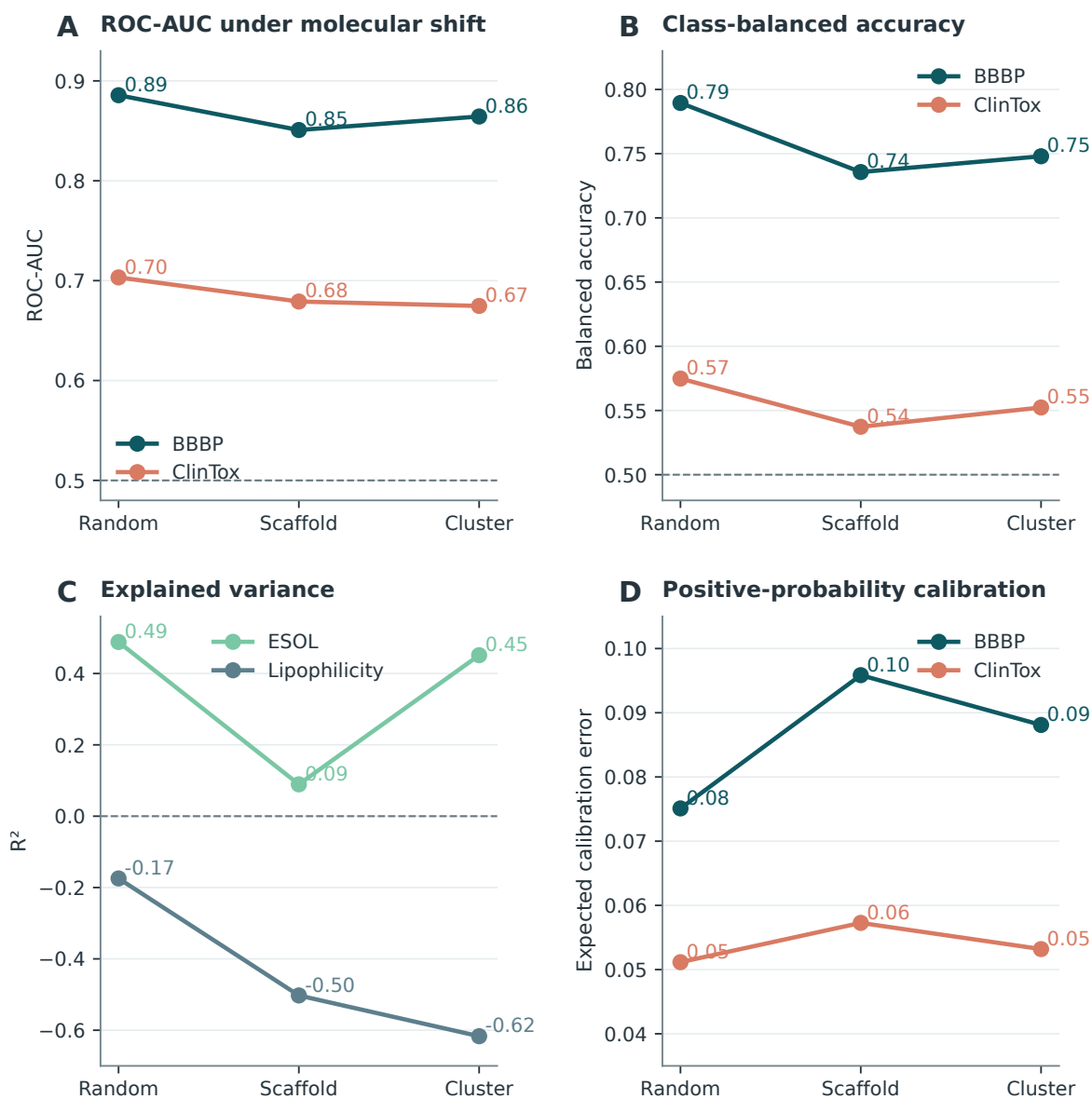
3.1 Molecular distribution shift changed point performance and calibration

Random splitting generally produced the strongest point performance, but the size and form of degradation depended on the endpoint (Figure 2). BBBP mean ROC–AUC declined from 0.886 under random splitting to 0.851 under scaffold and 0.864 under similarity-cluster splitting. Balanced accuracy decreased from 0.789 to 0.736 and 0.748, respectively. ClinTox was more difficult: ROC–AUC ranged from 0.675 to 0.703 and balanced accuracy from 0.537 to 0.575, with the random split again strongest.

The two regression endpoints separated a split effect from a weak-model effect. ESOL mean R^2 was 0.488 under random splitting and 0.451 under cluster splitting but fell to 0.089 under scaffold separation; RMSE concurrently increased from 1.441 to 2.035. Lipophilicity

mean R^2 was already negative under random splitting (-0.175) and deteriorated under scaffold (-0.503) and cluster (-0.617) designs. Conformal coverage close to 0.90 for this endpoint therefore could not be read as evidence of useful point prediction.

Positive-probability calibration also favored random splitting within endpoint. BBBP expected calibration error increased from 0.075 under random splitting to 0.096 under scaffold splitting; the corresponding ClinTox values were 0.051 and 0.057. Brier score and negative log-likelihood showed the same within-endpoint direction. Their smaller raw ClinTox values were not interpreted across endpoints because both scores depend on prevalence. Overall, the structured splits did not produce one uniform penalty: they changed discrimination, explained variance, and calibration in endpoint-specific ways.



Points are descriptive means across frozen model/regime combinations; model-specific cross-seed uncertainty is reported in the Supporting Information.

Figure 2: Point prediction and calibration under molecular distribution shift. Descriptive means across frozen model or model–regime combinations are shown for random, label-blind scaffold, and label-blind similarity-cluster splits. (A) ROC–AUC for BBBP and ClinTox. (B) Balanced accuracy. (C) R^2 for ESOL and Lipophilicity. (D) Positive-probability expected calibration error. Dashed lines denote chance discrimination or zero explained variance where applicable. Model-specific cross-seed uncertainty is reported in the Supporting Information.

3.2 Aggregate conformal coverage concealed class failure and low information

At nominal 90% coverage, overall classification coverage was close to target while class-specific behavior diverged sharply (Figure 3A,B). For BBBP, marginal overall coverage was 0.904–0.907 across splits, but positive coverage was 0.971–0.978 and negative coverage only 0.664–0.694. Density-ratio weighting changed these values only modestly. Mondrian calibration reduced the gap, although mean set size increased by approximately 0.208–0.232 and ambiguity increased by approximately 0.207–0.232.

ClinTox exposed the more consequential failure. Marginal overall coverage was 0.897–0.902, yet positive-class coverage was 0.068 under scaffold, 0.069 under cluster, and 0.099 under random splitting; negative-class coverage was 0.954–0.968. Shift weighting left overall coverage at 0.890–0.912 while positive coverage remained only 0.073–0.158. These sets looked precise rather than conservatively broad: mean size was below one because approximately 3.5–5.0% were empty, and two-label sets were rare.

Mondrian calibration raised ClinTox positive coverage by 0.841–0.872 relative to marginal prediction, with all eight frozen model–regime combinations showing model-specific cross-seed intervals entirely above zero. Positive coverage reached approximately 0.926–0.940 and negative coverage approximately 0.895–0.900. The repair was informative about uncertainty but inefficient as a binary decision: 70.9–71.9% of test compounds received $\{0, 1\}$, and mean set size reached 1.71–1.72. Thus overall coverage alone obscured two opposite defects—rare-class undercoverage for marginal sets and low decisiveness for class-conditional sets.

The same distinction reappeared during public Tox21 development (Figure 3C). ECFP density weighting produced minimum-class coverage of 1.00 across the 30 primary endpoint–seed cells, but MacroCSY was 0 and ambiguity was 1.00: every output was a two-label set. The physicochemical, score-view, ordinary, and TAME summaries occupied a lower-coverage but substantially more informative region. This was the empirical reason that a transport view could not be admitted merely because it increased coverage.

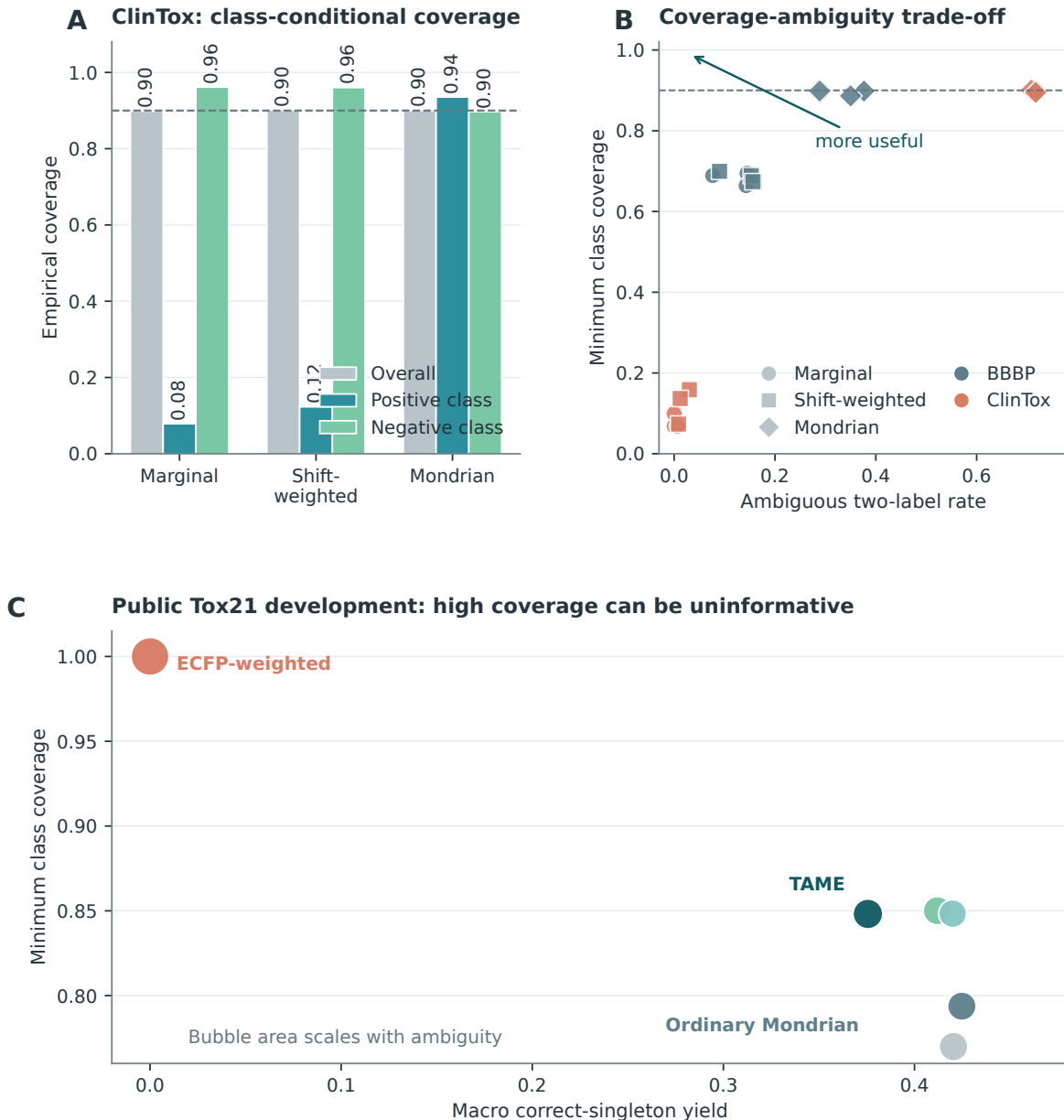


Figure 3: **Aggregate validity can hide either class failure or uninformative prediction sets.** (A) ClinTox overall, positive-class, and negative-class coverage for marginal, ECFP shift-weighted, and Mondrian conformal prediction, averaged across the frozen audit designs. (B) Minimum class coverage versus ambiguous two-label rate across BBBP and ClinTox split designs; marker shape identifies the conformal method and color identifies the endpoint. The arrow indicates the direction of jointly greater coverage and lower ambiguity. (C) Public Tox21 development summaries across six primary endpoints and five seeds. Bubble area is proportional to ambiguity; the ECFP-weighted comparator attained complete coverage by returning two labels for every compound.

3.3 Regression conformal variants did not show a universal advantage

Marginal, shift-weighted, and adaptive normalized regression intervals all produced coverage near 0.90, but neither added method consistently improved coverage and efficiency together. ESOL marginal coverage was 0.901 under random, 0.891 under scaffold, and 0.911 under cluster splitting. Paired shift-weighted and adaptive changes relative to marginal prediction were small (-0.010 to $+0.006$), and no comparison was supported across all four regression models. For Lipophilicity, shift weighting increased descriptive coverage by 0.006 under scaffold and 0.011 under cluster splitting, while increasing mean width by 0.174 and 0.234.

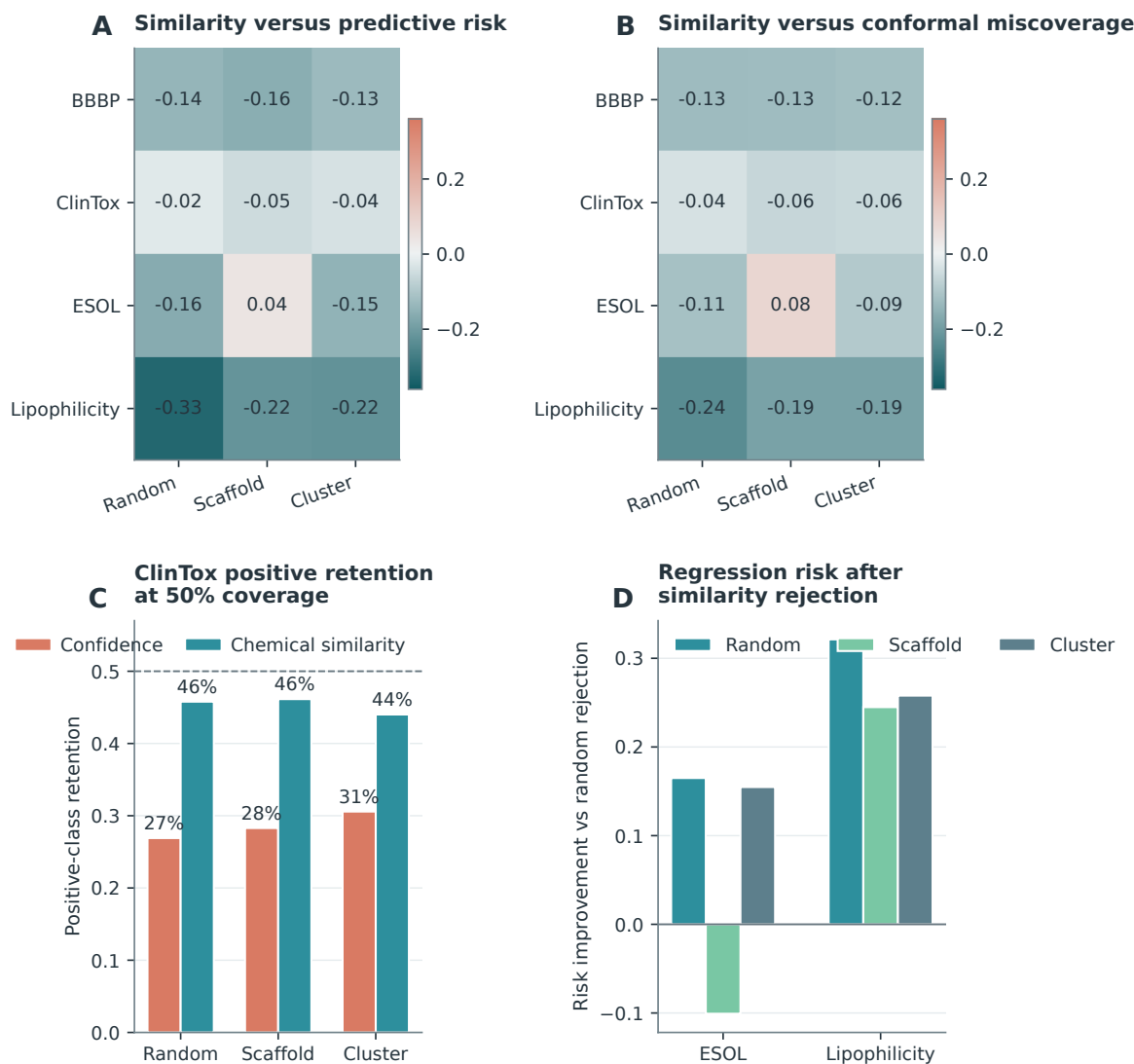
Adaptive normalization produced only modest interval heterogeneity: width coefficients of variation were 0.007–0.013 for ESOL and approximately 0.032–0.033 for Lipophilicity. Mean Spearman associations between interval width and realized error were weak, approximately 0.05–0.20. The audit therefore supplied no basis for assuming that estimated shift correction or local scaling would improve reliability across endpoints.

3.4 Chemical similarity and selective rejection were endpoint dependent

Maximum Tanimoto similarity to the training set ranked risk for some endpoints but not others (Figure 4A,B). BBBP similarity–risk correlations ranged from -0.131 to -0.159 and similarity–miscoverage correlations from -0.120 to -0.127 . Lipophilicity showed the strongest associations: approximately -0.22 to -0.33 for risk and -0.19 to -0.24 for miscoverage. ClinTox correlations were weak, approximately -0.02 to -0.06 . ESOL reversed under scaffold splitting, with correlations of approximately $+0.04$ for risk and $+0.08$ for miscoverage even though the random and cluster associations were negative. A fixed similarity threshold would therefore define different scientific contrasts across endpoint and split.

Selective prediction clarified why this mattered. At 50% overall retention, confidence-based ClinTox rejection retained only 26.9–30.6% of positive examples, whereas similarity-based rejection retained 44.0–46.1% (Figure 4C). Confidence lowered ordinary-risk partial area by 0.021–0.028 but worsened balanced-risk partial area by 0.021–0.029. The apparent benefit partly reflected a change in who remained.

Similarity-based regression rejection was likewise not transferable as a universal rule (Figure 4D). It improved risk at 50% retention for Lipophilicity by approximately 0.245–0.321 across all splits. For ESOL, it helped under random and cluster splitting but was worse than matched random rejection under scaffold splitting by approximately 0.101. Together, these results made two requirements explicit for Stage II: transport evidence had to be view- and batch-specific, and a method could not earn confidence by deleting baseline labels.



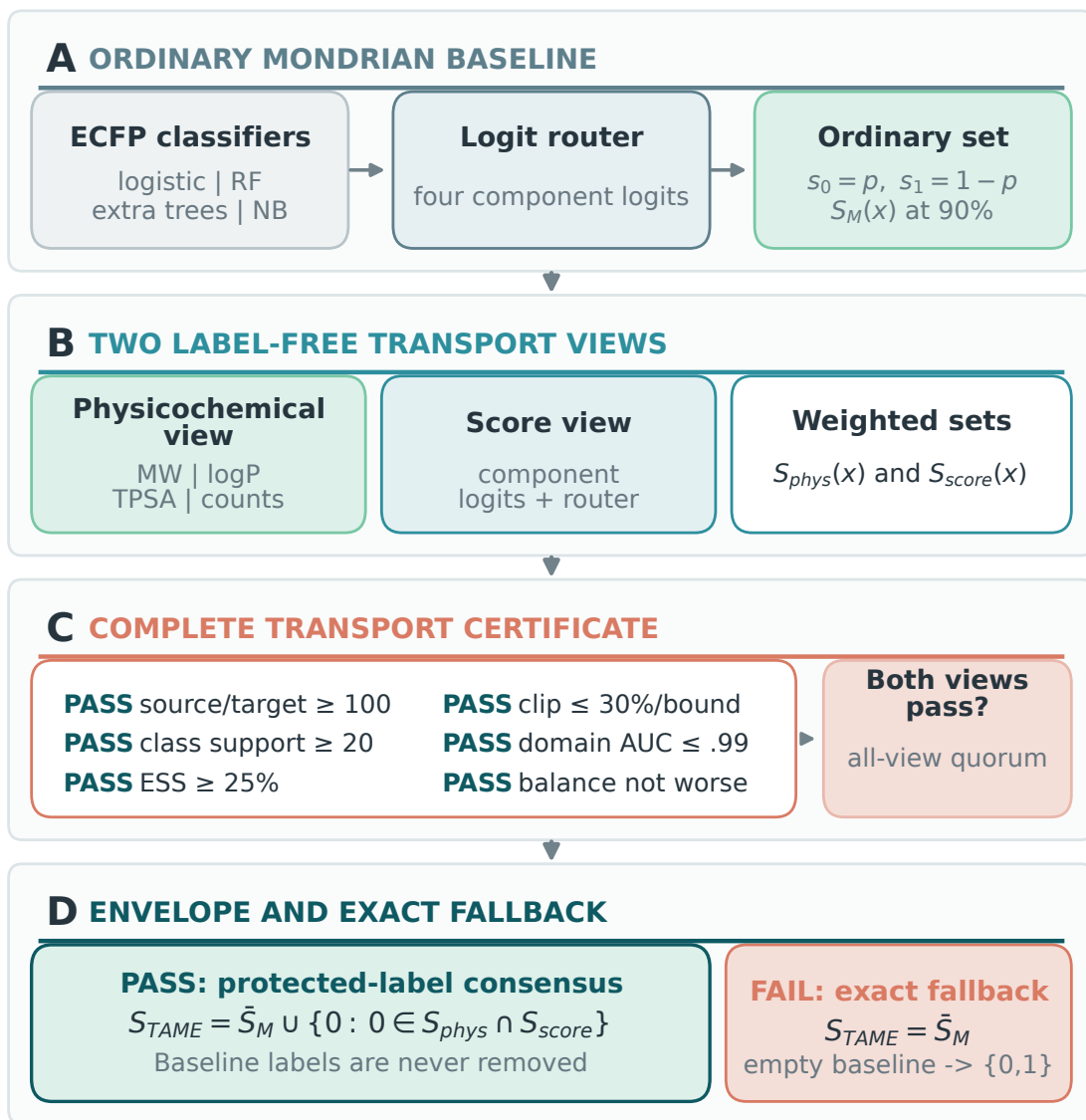
Negative heat-map values indicate increasing risk or miscoverage as similarity decreases.
Threshold partitions are sensitivity analyses, not universal chemical-domain rules.

Figure 4: Chemical-domain and selective-retention diagnostics are endpoint dependent. (A) Mean Spearman association between maximum training-set similarity and predictive risk. (B) Association between similarity and marginal conformal miscoverage. Negative values indicate increasing risk or miscoverage as similarity decreases. (C) ClinTox positive-class retention when 50% of all samples remain after confidence- or similarity-based rejection; the dashed line is composition-neutral 50% retention. (D) Improvement in regression risk at 50% retention for similarity rejection relative to matched random rejection. Threshold and rejection analyses are sensitivity diagnostics, not universal chemical-domain rules.

3.5 Audit failures became frozen TAME constraints

The audit did not identify a single scalar uncertainty score to optimize. Instead, it identified constraints: protect the weakest class explicitly; assess coverage jointly with correct-singleton yield; require batch-specific evidence before applying estimated transport weights; preserve every ordinary baseline label; expose an exact fallback; and separate architecture development from independent labels. Figure 5 shows how these constraints were encoded in TAME.

Development results were retained as design evidence only. Across 30 public primary cells, the frozen candidate increased mean class-0 coverage by 5.4394 percentage points, left mean class-1 coverage unchanged, and changed MacroCSY by -4.9169 points. It produced zero baseline-inclusion violations, zero empty sets, and zero newly created singletons. The score view passed in 30/30 cells and the physicochemical view in 24/30, meeting the 80% all-view activity gate; exact ordinary fallback handled the remaining six. The ECFP diagnostic failed in 30/30 cells because of effective-sample-size and/or clipping failures and was excluded from the envelope without relaxing its thresholds. All eight frozen promotion checks passed before any final EPA label was acquired or parsed.



$S_{TAME} \supseteq S_M$ | no empty sets | no newly created singletons
 Only singleton-to-defer transitions are permitted

Figure 5: **Mechanism and invariants of the transport-audited multi-view envelope.** (A) Four frozen ECFP classifiers feed a logit router and ordinary class-conditional Mondrian baseline. (B) Physicochemical and score views estimate separate label-free source-to-target weights. (C) Each view must pass the complete frozen transport certificate; the envelope requires an all-view quorum. (D) If both views pass, label 0 is added only by consensus while every baseline label is preserved. If either view fails, TAME returns the repaired ordinary baseline. The construction emits no empty set and cannot create a new confident singleton.

3.6 Sealed EPA validation improved minimum-class coverage within the efficiency guardrail

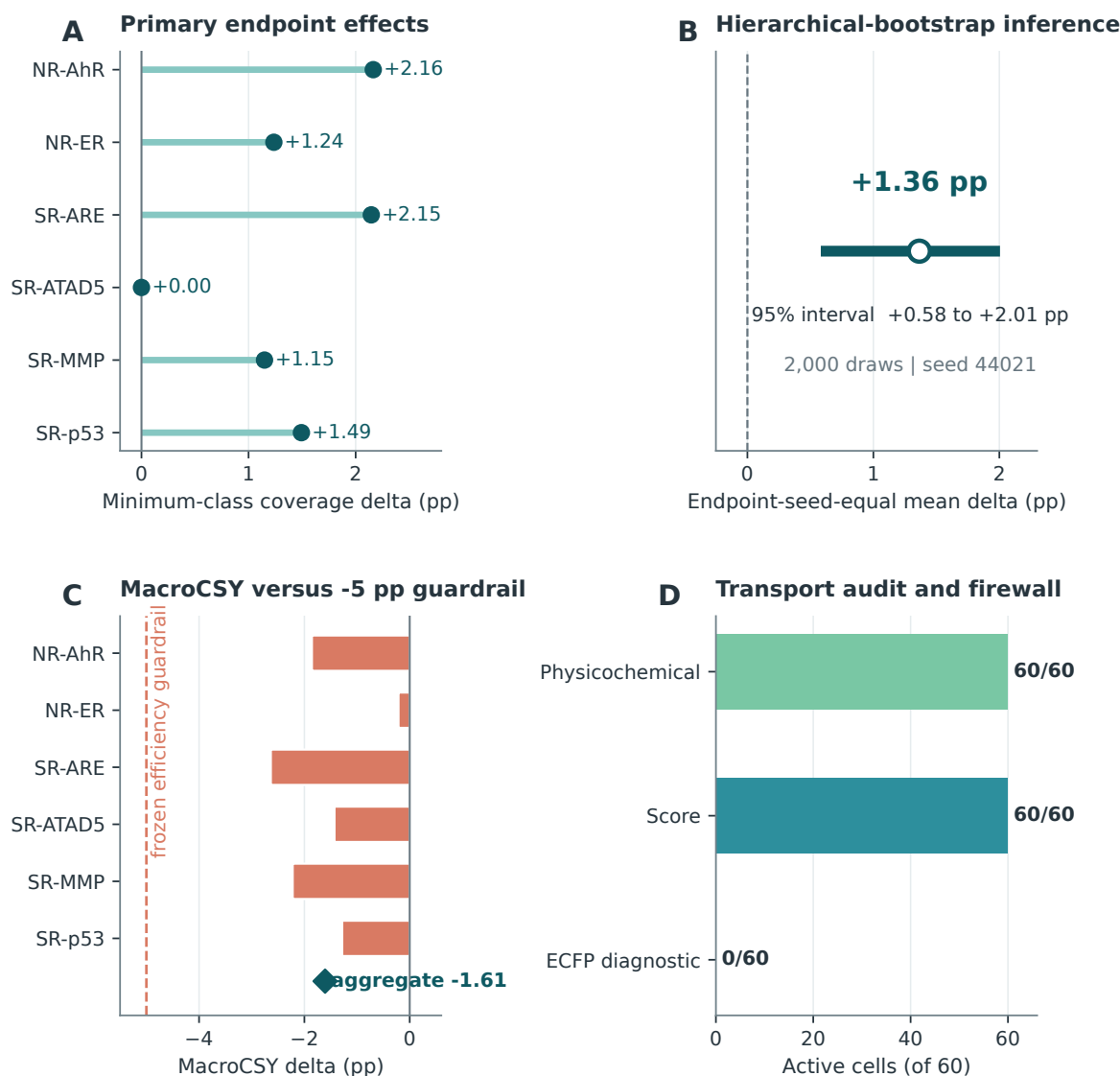
The independent run completed all 12 official endpoints across five fresh seeds (60/60 endpoint–seed cells); the primary analysis comprised the prespecified six endpoints and 30 cells. Two of 647 external structures failed the frozen standardizer and were retained as explicit full-set exclusions. Endpoint-specific standardized overlaps with the Tox21 training cohort were removed according to the lock. Predictions, the label-blind transport audit, and their hashes were written before the final label file was opened.

TAME increased the endpoint–seed-equal mean minimum-class coverage by 1.3649 percentage points relative to ordinary routed Mondrian prediction (Figure 6A,B). The deterministic 2000-draw hierarchical-bootstrap 95% interval was +0.5827 to +2.0051 points. Five of six endpoint means were positive: NR-AhR +2.1644, NR-ER +1.2362, SR-ARE +2.1461, SR-MMP +1.1489, and SR-p53 +1.4936 points. SR-ATAD5 had zero minimum-class-coverage change even though its class-0 coverage increased by 1.4363 points; the unchanged class remained the minimum.

The primary mean MacroCSY change was -1.6067 percentage points, inside the frozen -5 -point efficiency guardrail (Figure 6C). Because the envelope only adds a label, this efficiency cost is the expected counterpart of converting some singleton predictions into explicit deferrals. Structural checks confirmed the predeclared invariants: every TAME set contained its ordinary baseline set, no TAME set was empty, and no new singleton was created. Accordingly, wrong-singleton exposure did not increase on the same rows.

Both approved transport views passed in 60/60 final endpoint–seed cells, whereas the high-dimensional ECFP diagnostic failed in 60/60 because its frozen effective-sample-size, clipping, and balance certificate was not satisfied (Figure 6D). The diagnostic comparator nevertheless attained mean minimum-class coverage of 0.998 across all 12 endpoints while its mean ambiguity was 0.997 and MacroCSY only 0.002. This independent secondary result reproduced the central audit lesson: a coverage number can be nearly perfect because the output is nearly always uninformative.

The machine-readable predeclared result code was `coverage_gain_with_efficiency_noninferiority`. In publication-facing statistical language, the result is a coverage gain with the MacroCSY point estimate inside the prespecified efficiency guardrail, not formal evidence of non-inferiority and not an automatic declaration of scientific superiority. The result does not establish conditional coverage, exact validity for estimated density ratios, transfer to arbitrary molecular distribution shifts, or a general rule that label 0 should be protected in other assay families.



Firewall PASS: predictions and transport audit hashed before labels | 60/60 cells completed
Integrity PASS: deterministic interval repair | 96/96 tests passed

Figure 6: **Independent EPA validation under the prediction-to-label firewall.** (A) Mean TAME-minus-ordinary change in minimum-class coverage for each of the six prespecified primary endpoints across five fresh seeds. (B) Primary endpoint-seed-equal estimate and deterministic hierarchical-bootstrap 95% interval (2000 draws; seed 44021). (C) Endpoint-level MacroCSY changes and their aggregate; the dashed line is the frozen -5 -percentage-point point-estimate efficiency guardrail. (D) Final activity of the two approved label-free transport views and the ECFP diagnostic across all 12 endpoints and five seeds. Predictions and transport audits were hashed before labels were opened.

4 Discussion

This study began with a diagnostic question and ended with a constrained method. The diagnostic stage showed that point performance, probability calibration, chemical similarity, conformal coverage, set informativeness, and selective risk were not interchangeable measures of trust. The design stage then used those disagreements as requirements rather than averaging them away. Under the locked Tox21 evaluation, the resulting TAME envelope improved the weakest-class coverage by 1.36 percentage points on average, with a hierarchical-bootstrap interval above zero, while the 1.61-point MacroCSY loss remained inside the frozen 5-point point-estimate guardrail. Because final predictions were sealed before labels were opened, this is independent evidence for the prespecified candidate in this assay family rather than a retrospective selection from favorable endpoint–method combinations.

4.1 From a reliability illusion to an auditable intervention

The ClinTox audit provides the clearest example of an aggregate reliability illusion. Marginal and shift-weighted conformal prediction appeared close to the nominal 0.90 target overall while positive-class coverage remained between 0.068 and 0.158. Mondrian calibration repaired the class deficit but returned both labels for approximately 71% of compounds. Neither result is adequately summarized by *coverage near 90%*. The first is precise-looking but strongly asymmetric; the second is class-valid in the observed sample but often non-decisive. This distinction follows conformal work separating marginal from subgroup or class-conditional behavior and coverage from precision^{28–32}.

TAME does not attempt to erase that trade-off. It makes a one-sided, testable intervention: retain every ordinary Mondrian label, and add the development-identified protected label only when two audited transport views agree. The resulting invariants are stronger than a favorable empirical average. On any fixed prediction rows, class coverage cannot decrease, wrong-singleton exposure cannot increase, no empty set is emitted, and a new confident singleton cannot appear. What remains empirical is whether enough relevant baseline singletons are converted to deferrals on a new target cohort to produce a useful weakest-class gain without

an unacceptable loss of correct singletons. The sealed EPA evaluation directly tested that remaining question.

4.2 Transport diagnostics were decision gates, not decorative summaries

Weighted conformal prediction is theoretically motivated when the relevant likelihood ratio is known³³. In molecular applications the ratio is estimated in a selected representation, and that estimation step can dominate the result. In public development, the 2048-bit ECFP view produced complete coverage only by returning $\{0, 1\}$ for every target compound; it also failed its effective-sample-size and clipping certificate in every primary cell. In the final cohort it again failed all 60 endpoint-seed audits. Relaxing thresholds after seeing this behavior would have converted an observed failure into an unrecorded method change.

The lower-dimensional physicochemical and score views behaved differently. Their inputs encode complementary notions of shift: changes in interpretable molecular properties and changes in the fitted ensemble’s response geometry. Both passed in every final cell, while the public development physicochemical view failed in six cells and triggered exact fallback. This matters because a transport audit should be allowed to say *do not transport*. Domain-classifier AUC alone is insufficient; the effective sample size, boundary clipping, class support, and whether weighting actually improves multivariate balance all affect whether a finite-sample correction is usable. Molecular implementations such as CoDrug show the promise of density-aware conformal prediction²⁷; our results add evidence that a shift correction should expose its failure conditions rather than be applied automatically.

4.3 Relation to recent conformal methods under shift and efficiency constraints

TAME is closest in motivation to shift-aware conformal methods, but its estimand and intervention differ. KMM-CP seeks coverage correction through kernel mean matching and selective support overlap³⁶; TAME instead asks whether two lower-dimensional, label-free views have enough finite-sample support to permit a one-sided expansion of an ordinary

Mondrian set. Multi-distribution robust conformal prediction offers uniform validity when labeled calibration data are available from multiple source distributions³⁷. RACER-C4 supplies one labeled training source and one unlabeled external target cohort, so that guarantee is not available here.

The efficiency objective is also deliberately different. SOCOP redesigns nonconformity scores to increase singleton frequency³⁸, whereas TAME cannot increase singleton yield because it only adds a label. Batch conformal prediction pools evidence across multiple test examples to construct a joint set of label vectors³⁹; TAME uses batch covariates only to audit transport and still emits one binary set per compound. These methods are complementary rather than interchangeable baselines. A fair head-to-head comparison would need a new, prospectively locked protocol that harmonizes data access, output objects, and estimands; adding such experiments after the EPA labels were opened would compromise the present firewall.

4.4 Efficiency loss represents explicit deferral

MacroCSY was chosen because ordinary set size alone does not distinguish a useful singleton from a wrong one. For a true class, correct-singleton yield asks how often the method returns exactly the correct label; averaging equally across classes prevents prevalence from dominating this efficiency measure. TAME can only preserve a singleton or turn it into a two-label set, so its MacroCSY cannot improve by construction. The nontrivial scientific question was therefore whether the sealed cohort would show a measurable weakest-class gain while the MacroCSY point estimate remained inside the guardrail fixed before independent labels.

The final 1.61-point loss was inside the 5-point guardrail, and every primary endpoint mean lay on the accepted side of that boundary descriptively. The threshold means that no more than five correct singleton outputs per 100 class-balanced cases could be sacrificed at the aggregate point-estimate level. It was an architecture-selection tolerance, not a clinically validated utility margin, and no confidence-interval-based non-inferiority analysis for MacroCSY was frozen. This result should therefore be read as a measured deferral cost inside a prespecified guardrail, not as a free gain or a formal non-inferiority finding. Whether the trade is desirable

in a particular screening program depends on endpoint-specific costs of false exclusion, false inclusion, ambiguity, and experimental follow-up. The present study did not assign clinical or regulatory utility to those outcomes.

4.5 Chemical similarity remains useful but cannot serve as a universal trust score

The Stage I applicability-domain results explain why ECFP novelty was retained as a diagnostic rather than promoted to a universal gate. Similarity ranked risk and miscoverage for BBBP and especially Lipophilicity, was weak for ClinTox, and reversed for ESOL under scaffold shift. A molecule can be structurally novel in one representation yet predictable because the endpoint depends on features shared across scaffolds; a close analogue can also be difficult in a steep or noisy local structure–property region. This endpoint dependence agrees with recent molecular uncertainty benchmarks and distance-based analyses^{13;14;16;19}.

Selective prediction supplied a related warning. Confidence-based ClinTox rejection lowered ordinary risk partly by retaining only 27–31% of positives when half of all observations remained. Risk–coverage curves can therefore combine error ranking with a change in class composition. Molecular abstention studies should report balanced risk and class-specific retention, while applicability-domain studies should report continuous associations and threshold sensitivity. A fixed Tanimoto threshold is a representation-specific operating choice, not a chemical constant. This composition-aware interpretation is consistent with feature-aware predictive-inference work, which treats accepted-set behavior as part of the estimand rather than an incidental consequence of rejection⁴⁶.

4.6 Relation to contemporary ADMET model benchmarks

The present contribution is complementary to architecture-centered ADMET studies. Broad benchmarks map the predictive capabilities of foundation models, graph neural networks, AutoML systems, and classical algorithms under realistic data challenges⁴⁰. Our representation-controlled audit instead isolates conflicts among post-prediction reliability quantities; the

RACER-C4 protocol then tests the fixed TAME uncertainty layer with an independent-label firewall. The design cannot establish where TAME would rank when attached to another representation, but it can expose a failure mode that a point-performance leaderboard does not measure.

This separation is important for reproducibility. Stage I model families were never treated as independent replicates. Stage II primary endpoints were selected by a public label-count rule, not by final performance. Development and prospective seeds were disjoint. Null and adverse endpoint results were retained, including the zero SR-ATAD5 minimum-class change and every negative MacroCSY change. Finally, the deterministic bootstrap repair changed only traversal order in an inference-only calculation; predictions, labels, point estimates, and the frozen interpretation were unchanged and both reports remain versioned.

4.7 Limitations and scope

The diagnostic panel contained four public endpoints, an ECFP4 representation, and conventional model families. It was designed to expose contrasting failure modes, not to represent the full ADMET task universe. The independent evaluation was restricted to the Tox21 nuclear-receptor and stress-response assay family and to one released external cohort. It was not a prospective wet-laboratory experiment, a clinical validation, or evidence of regulatory utility. The primary interval is conditional on the six prespecified endpoints, five seeds, locked preprocessing, and observed final cohort.

TAME also has deliberate structural limits. It only adds a development-selected label, so it cannot improve correct-singleton yield and will not repair a deficit in an unprotected label. The choice of label 0 was specific to the public development batch and must not be transferred to another assay family without a new, prospectively separated development process. The two transport audits are empirical diagnostics rather than proofs that density-ratio assumptions hold. Passing them does not supply conditional coverage or exact validity under arbitrary shift; failing them only certifies that the frozen augmentation is disabled.

Some external endpoint rows were excluded because standardized structures overlapped

the training cohort, and two structures could not be parsed under the locked standardizer. Those exclusions were defined before labels and retained by identity, but they reduce the effective external sample. Rare classes also limit the resolution of endpoint-specific coverage estimates. Broader evaluation should include new assay families, temporal cohorts, learned molecular representations, and prospectively generated experimental data while preserving the same separation between design and final labels. Assay labels can also contain biological and measurement heterogeneity that no structural split or conformal wrapper removes; robustness of marginal coverage to label noise does not by itself resolve the scientific meaning of the target⁴⁷.

4.8 Conclusion

Reliable molecular prediction cannot be reduced to aggregate accuracy, marginal coverage, or a single distance-to-training score. The four-endpoint audit showed how each can fail: structured splits changed performance, marginal coverage concealed class undercoverage, class-conditional repair could become ambiguous, similarity was endpoint dependent, and selective rejection could change who remained. TAME converted those observations into an auditable intervention with explicit transport certificates, exact fallback, baseline containment, and a prediction-to-label firewall. In the locked EPA evaluation it improved minimum-class coverage while its MacroCSY point estimate remained within the prespecified efficiency guardrail. The defensible conclusion is narrow but useful: failure-aware constraints and sealed validation can improve the credibility of a molecular uncertainty layer, while generalization beyond the evaluated Tox21 setting remains an empirical question.

Data and Code Availability

All datasets analyzed in this study are public molecular-property datasets. The preprocessing, label-blind split generation, model training, calibration, conformal prediction, applicability-domain analysis, selective-prediction analysis, RACER-C4 executable lock, promotion record, deterministic inference repair, integrity audits, frozen result tables, manuscript figures, and

LaTeX source files are maintained in the [public reproducibility repository](#). The repository includes versioned source code, cryptographic manifests, frozen manuscript-ready result tables, and instructions for rebuilding the manuscript and Supporting Information.

Declaration of Competing Interest

The authors declare no competing interests.

CRedit Authorship Contribution Statement

Siyuan Tong: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing. Yuechen Wang: Methodology, Validation, Writing – review and editing. Both authors approved the final manuscript.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI ChatGPT to support language editing, manuscript organization, LaTeX formatting, code review, and reproducibility-material preparation. After using this tool, the authors reviewed and edited all content as needed, verified the numerical results, code, figures, and references, and take full responsibility for the content of the submitted manuscript.

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